

Demo: Open Visual Positioning and Discovery Services for Location-based Augmented Reality

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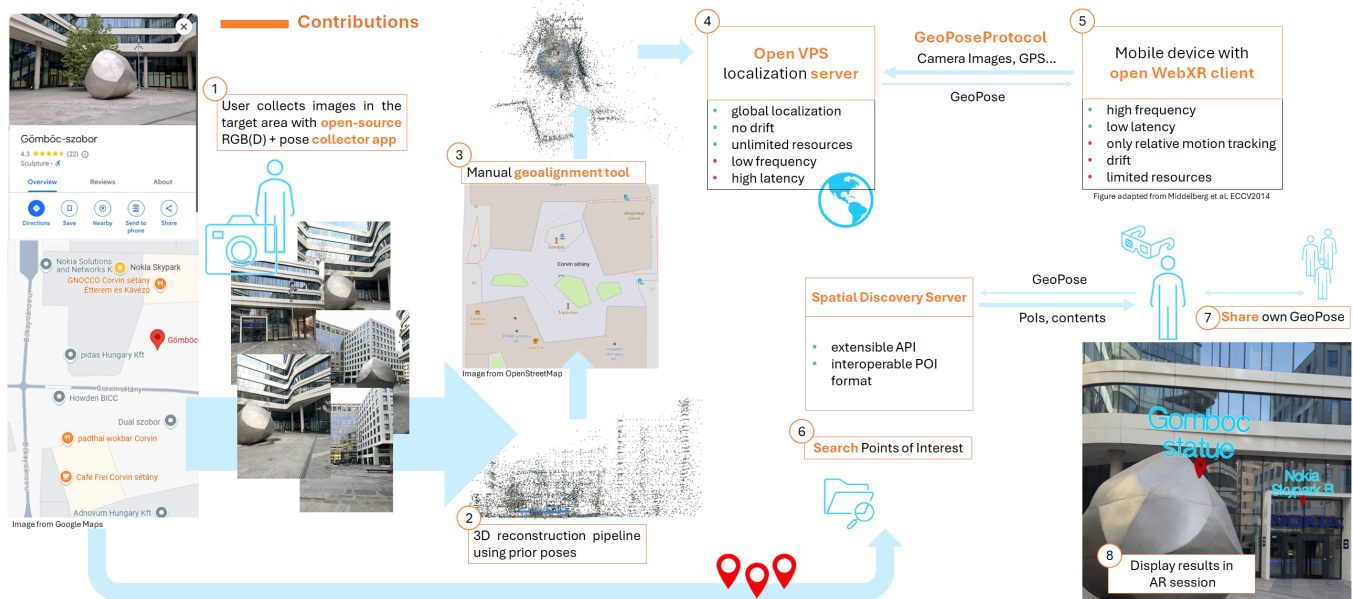


Figure 1: (1) A user begins by taking photos of the target area. (2) These are fed to a 3D map reconstruction pipeline. (3) The 3D map is aligned with the geodetic coordinate system. (4) Our OpenVPS determines the GeoPose of the user's AR device based on (5) one or more images. (6) The AR device queries nearby POIs and (8) renders those within the user's field of view in the AR session. (7) The GeoPose of each user is shared with the other users. Video supplement: <https://youtu.be/JTqQ0HALyXY>

Abstract

We present a key milestone towards open and interoperable location-based augmented reality (AR). We demonstrate our prototype open visual positioning service that can determine the geopose of AR devices, examples of geospatial digital content retrieval around the physical location of the devices, and collaboration enabled by sharing device poses and virtual objects in universal geodetic coordinates.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; Mixed / augmented reality; Collaborative and social computing systems and tools; • **Information systems** → Global positioning systems; • **Networks** → Location based services.

Keywords

Mixed/augmented reality, Ubiquitous and mobile computing systems and tools, Location based services

ACM Reference Format:

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1 Introduction

Geospatial augmented reality (AR), the AR cloud, the spatial Web, and spatial computing in general build upon persistent, location-specific digital content that can be discovered and consumed in context at a relevant physical location. The use cases for these enablers include information in an urban environment about commerce, transportation, navigation, restaurants, maintenance, entertainment opportunities and other points of interest appearing in the user's field of view as if attached to places and objects.

Geospatial AR cloud relies on technologies including 3D mapping, precise localization, content discovery, reality modeling, and the rendering of personalized content to users in AR sessions. Today, these technologies are provided in multiple vertically integrated proprietary platforms. For true multi-agent and multi-vendor AR experiences, there must be ways of sharing AR assets based on user location and poses outside of closed technology silos. The Open AR Cloud community has been contributing to open and interoperable components towards an open spatial computing platform (OSCP) [2], which we also build upon.

2 Related Work

To make digital objects appear at the same location for all users at any time, continuous and precise user localization both in position and orientation (6 degrees of freedom) is required. These requirements can be fulfilled with global localization via a visual positioning system (VPS) combined with local egomotion tracking via visual-inertial odometry (VIO). Proprietary solutions are widely available for motion tracking for AR (ARKit, ARCore), for anchoring digital content to objects or places (Vuforia, Azure Spatial Anchors, Magic Leap Anchors) and even to geographic locations (Niantic Lightship, Google Geospatial Anchors, Apple GeoAnchors). Unfortunately, these frameworks and enablers do not communicate with one another and cannot be mixed in the same AR experience.

While closed commercial solutions may deliver faster and more accurate performance, their coverage may not be continuous and, when a user wants to cover a new area, developers and users have no influence on the mapping process or conditions, or where and when a map gets captured or updated. Niantic¹ gamifies the mapping experience and a user can capture their own map, but the map's specific features are unknown, and the user does not control the captured map. Immersal² and Augmented City³ provide tools for community mapping and map management, but the maps are proprietary and cannot be combined with maps of other providers. These constraints significantly hinder the wider adoption of geospatial AR based on VPS.

We believe that openness, interoperability, and crowdsourcing are required to create and maintain the large scale of maps required

for ubiquitous geospatial AR experiences. Similar to the community-maintained 2D OpenStreetMap, the platforms Mapillary, OpenDroneMap, and KartaView provide(d) crowdsourcing tools for street imagery and 3D mapping, but these maps are not meant for AR device localization. The scientific literature on 3D reconstruction and collaborative mapping is vast, but most researchers have focused on the algorithmic aspects rather than enabling interoperability that, in and of itself, increases innovation and ease of use.

We are building and demonstrating an open-source toolset that enables setting up a geospatial VPS and the continuous and interoperable discovery of location-based content for any AR experience. It is applicable for creation and delivery of AR experiences and serves as a valuable tool for research in AR and other adjacent location-based services.

3 An Open Visual Positioning System

Our first contribution is a simple but fully open-source visual positioning pipeline including tools and services for image capture, map creation, map geo-alignment, and localization⁴.

An open VPS should support swappable localization algorithms and map formats for R&D environments and testbeds. Research on localization algorithms is out of scope for this demo, instead we make use of robust and well-proven localization methods from the computer vision literature. We adopt the HLoc⁵ localization framework as the core, because it already supports a wide range of state-of-the-art methods and it is easy to add new options in the future. Using our extensible universal API GeoPoseProtocol⁶, we abstract the algorithms and leverage the OGC GeoPose⁷ standard data exchange protocol to encode location and pose coordinates.

Usually, expensive sensor rigs (Google car, BLK2GO, etc.) with Lidar and multiple cameras are used for 3D mapping, which are not available to normal developers and users. If simplicity is preferred over accuracy, cheap, ubiquitous smartphones and open structure from motion (SfM) algorithms can be used for mapping. For simplicity, we record sequences with the open StrayScanner⁸ application, but there are many other similar tools. SfM does not scale well with the size of the area, and we observed that reconstructions often converge to incorrect solutions and invalid 3D maps. Fortunately, we can use prior information from the pose tracking of the AR engines in smartphones to address the map scale ambiguity.

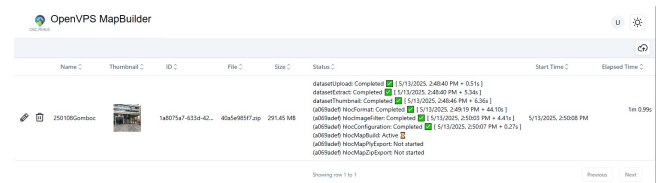


Figure 2: MapBuilder user interface for uploading data and managing maps.

⁴<https://github.com/OpenArCloud/openvps>

⁵<https://github.com/cvg/Hierarchical-Localization>

⁶<https://github.com/OpenArCloud/oscp-geopose-protocol>

⁷<https://www.ogc.org/standard/geopose/>

⁸<https://github.com/strayrobots/scanner>

¹<https://www.nianticspatial.com/>

²<https://immersal.com>

³<https://augmented.city/>

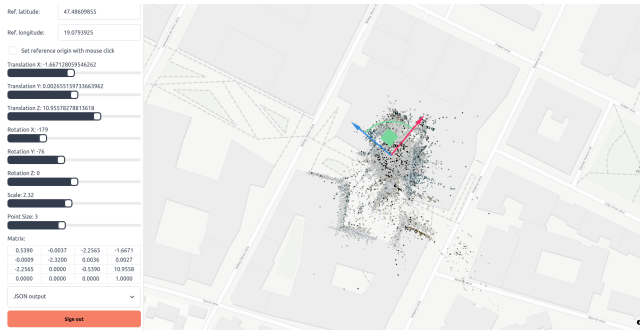


Figure 3: MapAligner user interface for georegistration.

Our system consists of three services: a MapBuilder, a MapAligner, and a MapLocalizer service. The mapper person can simply upload the recording from the capture smartphone to our MapBuilder service (see Figure 2) and start the automatic map creation. Once the map is created, it can be extracted in point cloud format, and the mapper person can align the 3D map with the world with a 2D satellite image or building floorplan manually (see Figure 3) in our MapAligner tool. Finally, the geo-aligned map can be deployed to the MapLocalizer service which implements the GeoPoseProtocol API to serve localization requests.

An end-user device sends a query image (or images, and potentially other sensor input) to the MapLocalizer that passes it to the core localization method(s). The query 2D visual features are matched with the 3D visual features in the spatial map to estimate the six-degrees-of-freedom pose of the camera in the 3D map, and then through our geo-registration transform indirectly in the world. The calculated GeoPose is sent back to the AR device which can then align its local coordinate system with the world and start searching for spatially anchored contents in its neighborhood.

4 Searching Points of Interest

Our second contribution is the continuous discovery of interoperable points of interest. The Open Geospatial Consortium⁹ is developing a new extensible standard (draft) for Point of Interest (PoI) descriptors. A PoI is a location about which information is available, it can be as simple as a set of coordinates and an identifier, or more complex such as a three-dimensional model of a building with names in various languages, information about open and closed hours, and a civic address. We adopt this PoI format as it enables any content publisher to effectively describe and efficiently serve and exchange PoI data. We have created a PoI provider and aggregator service and integrated it with our system. In our aggregator service, we currently use Google Maps and OpenStreetMap, but in the future, other PoI service providers can be added. Our service searches for nearby places based on the GeoPose and user-defined keywords (or categories like restaurant). The place descriptors are filled into OGC PoI format, duplicates are merged, and sent back to the client. As shown in Figure 1 and Figure 4 and in the supplemental video, we perform PoI content retrieval based on current location and present them interactively in mobile AR.

⁹<https://github.com/engeospatial/poi>

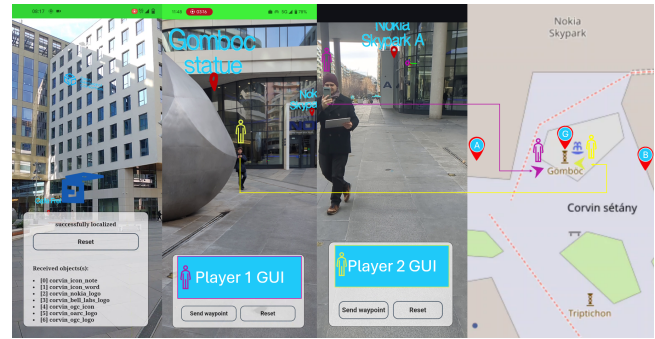


Figure 4: Multiple users share their GeoPoses with one another and they also appear in a live map view.

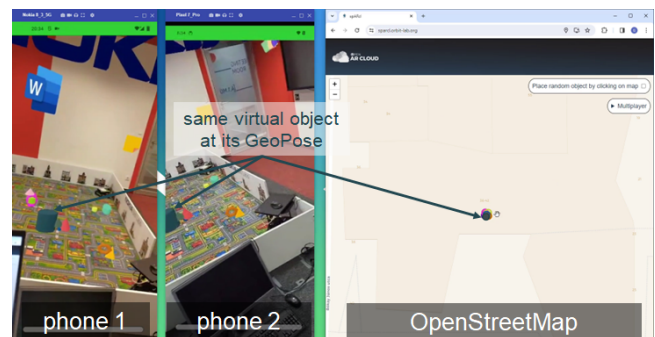


Figure 5: Shared geospatial AR: a user clicks on the floor to spawn a new virtual object at the GeoPose of the hitpoint. The new object is shared with other users and appears at the same physical place for everyone, even on a traditional map. The other way around, clicking on the 2D map also places new objects in AR the users' views.

5 Collaborative Applications

Our third contribution is to enable sharing the GeoPoses of users (Figure 4) and spawned digital objects (see Figure 5) through a pub-sub message broker. Once geo-localized at least once and tracking their egomotion, poses of users (and any dynamic objects) can be expressed as GeoPoses and published in the message broker. We also enable spawning new content by clicking on the ground via raycasting in the AR session and calculating the GeoPose of the hitpoint. Such a geo-location cursor can have many interesting future applications. Connecting non-AR viewers (like the OSM view in Figure 5) and authoring tools also requires unified handling of locations, hence there is need for such an interoperable way of sharing locations, digital objects, and actor poses for true multi-agent and multi-vendor AR experiences. We note that sharing the precise poses of users implies security and privacy challenges, but these are out of the scope of this work.

6 Integration with other Spatial Services

We integrated our Open VPS and our PoI Search with the service and content discovery mechanisms of the Open Spatial Computing Platform [2]. The interaction flow between a client device and

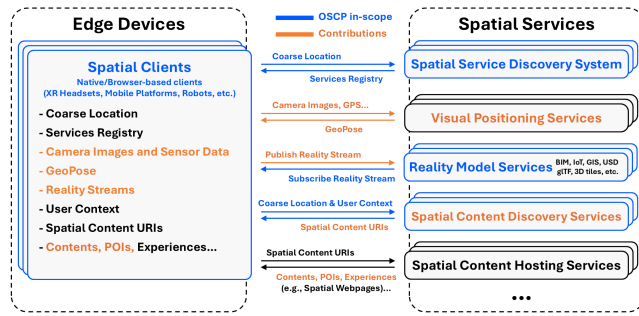


Figure 6: Overview of the client-platform interaction flow in the Open Spatial Computing Platform and our contributions highlighted in orange.

the platform is illustrated in Figure 6, with our new contributions highlighted. Spatial service discovery provides both a map registry and localization service registry. In the simplest case, a VPS covers only a single area, but in general, one service should cover many places and perform coarse localization based on auxiliary signals (GPS, WiFi SSID, etc.) to select candidate map(s), and then perform fine localization in the respective 3D map. In OSCP, all contents are tied to physical locations described by GeoPose (in other words to universal coordinates and not to a particular map using a particular proprietary technology).

In the future, we envision to extend our system with location-anchored, community-created digital media content, interactive visual navigation, restaurant discovery, lecture discovery, ride hailing, etc. applications to name only a few. The content can be as simple as text bubbles, Web URLs, directional arrows or labels, but also complex interactive experiences including 3D models and 3D sounds, all delivered in a Web App with contextual search. Furthermore, we plan to integrate location-based search for Internet-of-Things devices and automatically generated AR user interfaces for sensors and actuators, as well as robots.

7 Demonstration

We will demonstrate all elements of the above contributions, given that the experience can be quickly deployed at a new location. We will showcase (similar to Figure 1) the geo-localization of users via our open VPS, the discovery of local POIs, the creation of new objects by clicking on the ground, as well as multi-user interaction. There is no application install needed, our demo runs in an Android Chrome browser on any WebAR-capable device. Participants with deeper interest can also take a closer look at our Web-based map builder service and our geo-alignment tool. The demo can be set up both indoors and outdoors, and we plan a small indoor setup for the demo session, and a larger outdoor setup for a related tutorial session [1]. In our indoor demonstration, we will show a few hand-picked POIs that can be meaningful indoors, while outdoors we will use our search and aggregator service for finding POIs around the demo venue. The outdoor deployment and the eventual results of the tutorial’s hackathon can be enjoyed by the conference participants during the whole conference week.

8 Discussion and Impact

Our OpenVPS enables the precise localization of any camera device in a global shared coordinate system, and our extensible spatial search enables the continuous query of heterogeneous data sources. Unfortunately, without precise geodetic measurement devices, we have no means to quantify the accuracy of the OpenVPS at out test scenes. We have built the service around the HLoc framework which is also meant for benchmarking existing and future localization algorithms on public datasets, and the concrete methods are easily switchable in our service. In our experiments, the points of interest pins appear in AR in close vicinity (within 1 m error) of their expected geo-locations. The round-trip localization delay is about 3 seconds, which we consider slow, but it is sufficient to localize only once as long as the user is not moving too far away and the drift of the ego-motion estimation is small (see the supplementary video).

Enabling the rapid creation of VPS maps with a universal shared GeoPose coordinate system and an extensible unified API may have an important impact. Building large maps is difficult, but collaboratively building and updating many small maps and stitching them is easily scalable, so that we can cover the whole world similar to the small cells of the communication networks. We imagine in the future a large-scale visual positioning system with many public and private VPS providers and transitioning between them similar to roaming in cellular radio networks. Each cell may use a different mapping technology or different localization provider, but if the APIs are compatible, our proposed solution could even bridge existing proprietary solutions. In such a network of 3D maps, building AR experiences at large scale becomes significantly easier than today. Our solution currently enables quick setup, self-hosting, and experimentation at small scale. It is at the level of a research tool for now, but it may have high societal value in a future spatial Web.

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